Lesson 1 - 07/04/25 - Project Intro

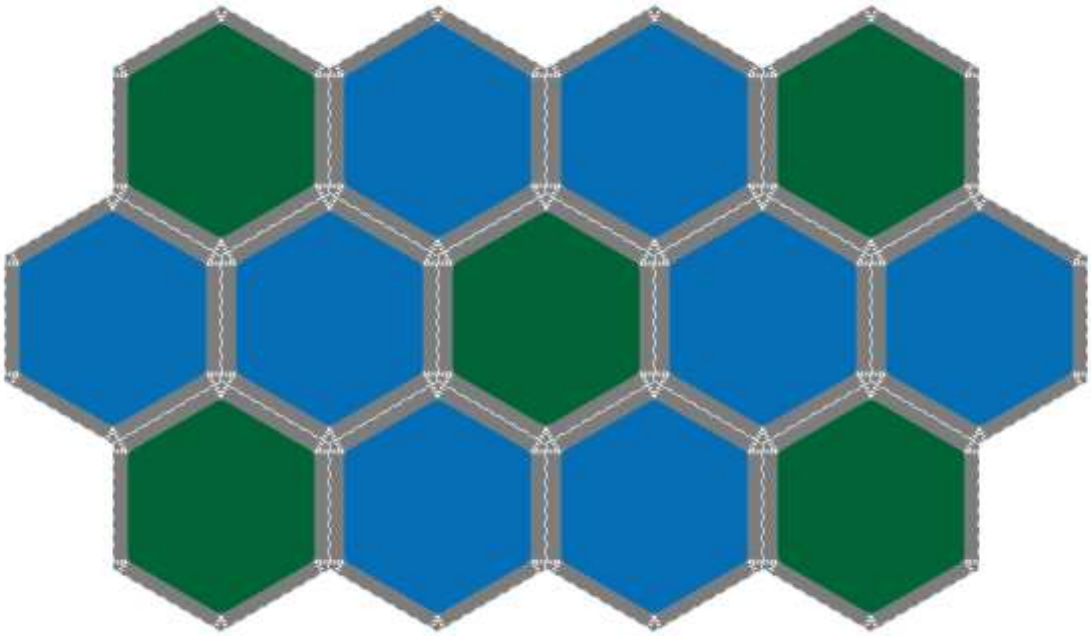
Publicado

Access your groups' OneNote LogBook in this link (http://bit.ly/hexcity2025)

Discover your Creative type here! (https://www.mycreativetype.com/)

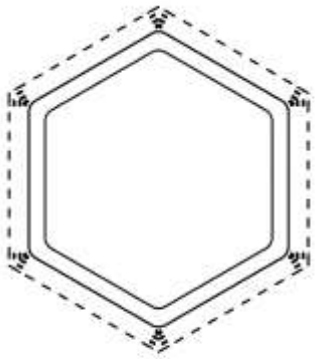
Project Objective

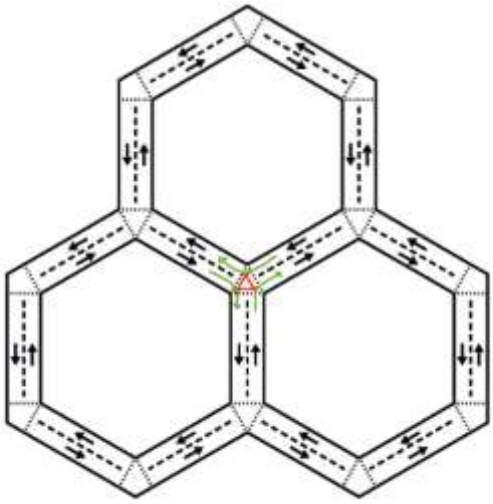
9th-grade students will design and build three independent smart city models (9A, 9B, 9C), each composed of 13 hexagonal wooden blocks -> 8 hexagonal wooden blocks (7 groups with 1 per group and 1 for presentation purpose, 30–35 cm) with Arduino-based solutions addressing São Paulo's urban challenges and 5 reserved green hexagons to ensure vegetation and sustainability. The focus is on SDG 7 (Clean Energy) and systemic thinking, emphasizing how interconnected solutions create sustainable cities.

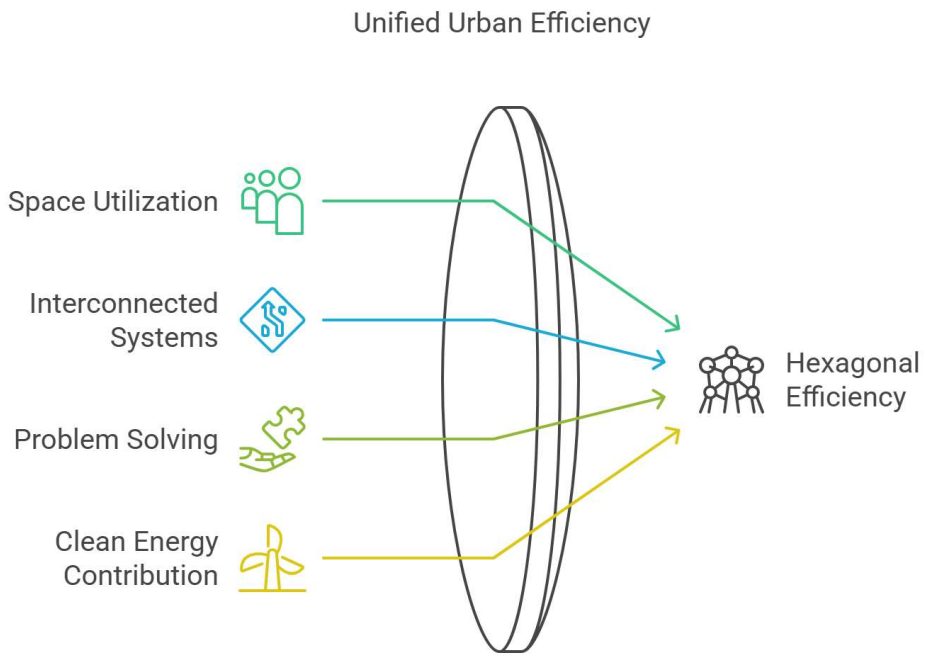


Core Principles

1. Hexagonal Efficiency:
   - Hexagons symbolize efficient space use and interconnected systems, even if not physically modular.
   - Each hexagon solves a problem while contributing to the city's clean energy ecosystem.
   - Maintains the ratio of each build.





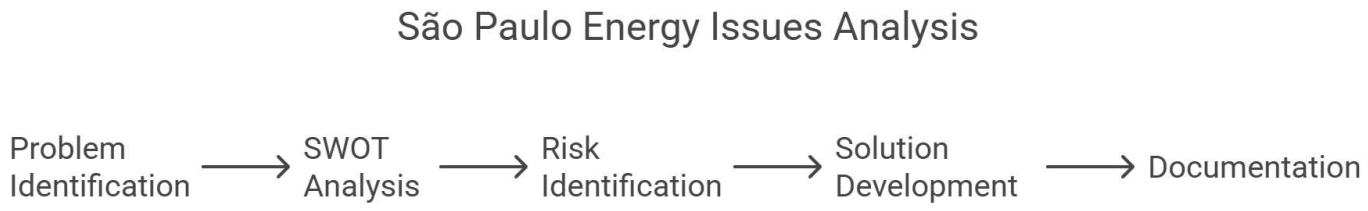


Phases of Work

1. Brainstorming & Risk Analysis

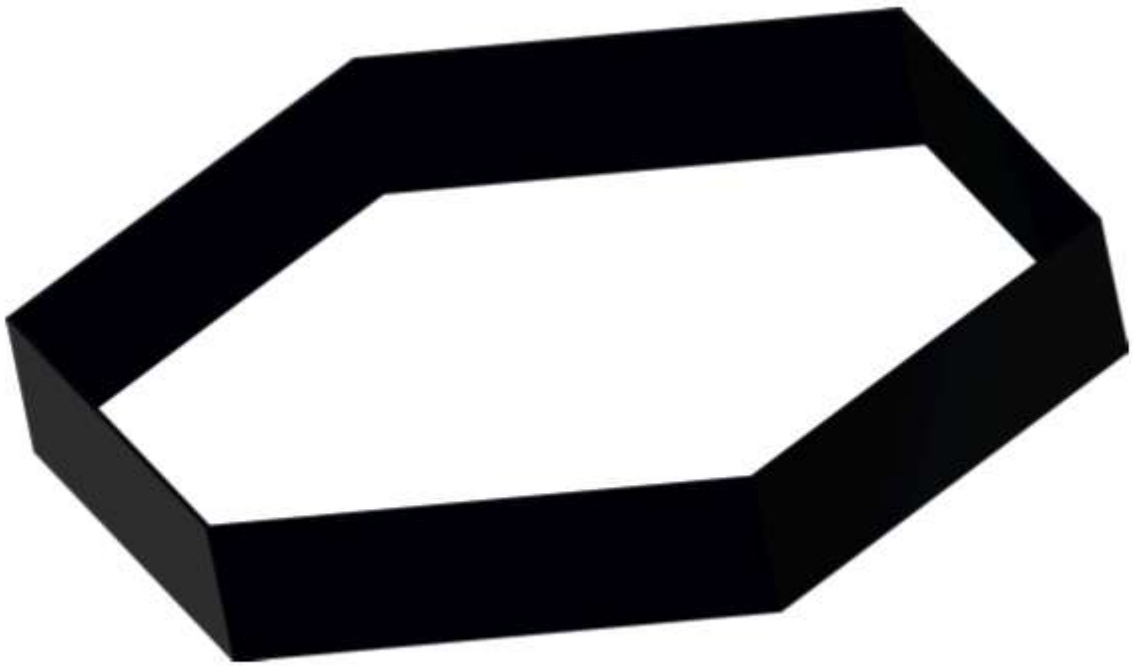
- Activity:

○ Problem Identification: Students brainstorm São Paulo's energy issues (blackouts, fossil-fuel reliance, etc).
○ Risk Assessment: Groups use SWOT analysis to identify risks ("Solar panel failure on cloudy days → Add battery backups").
• Tools:
○ OneNote Diary: Document ideas, risks, and solutions.



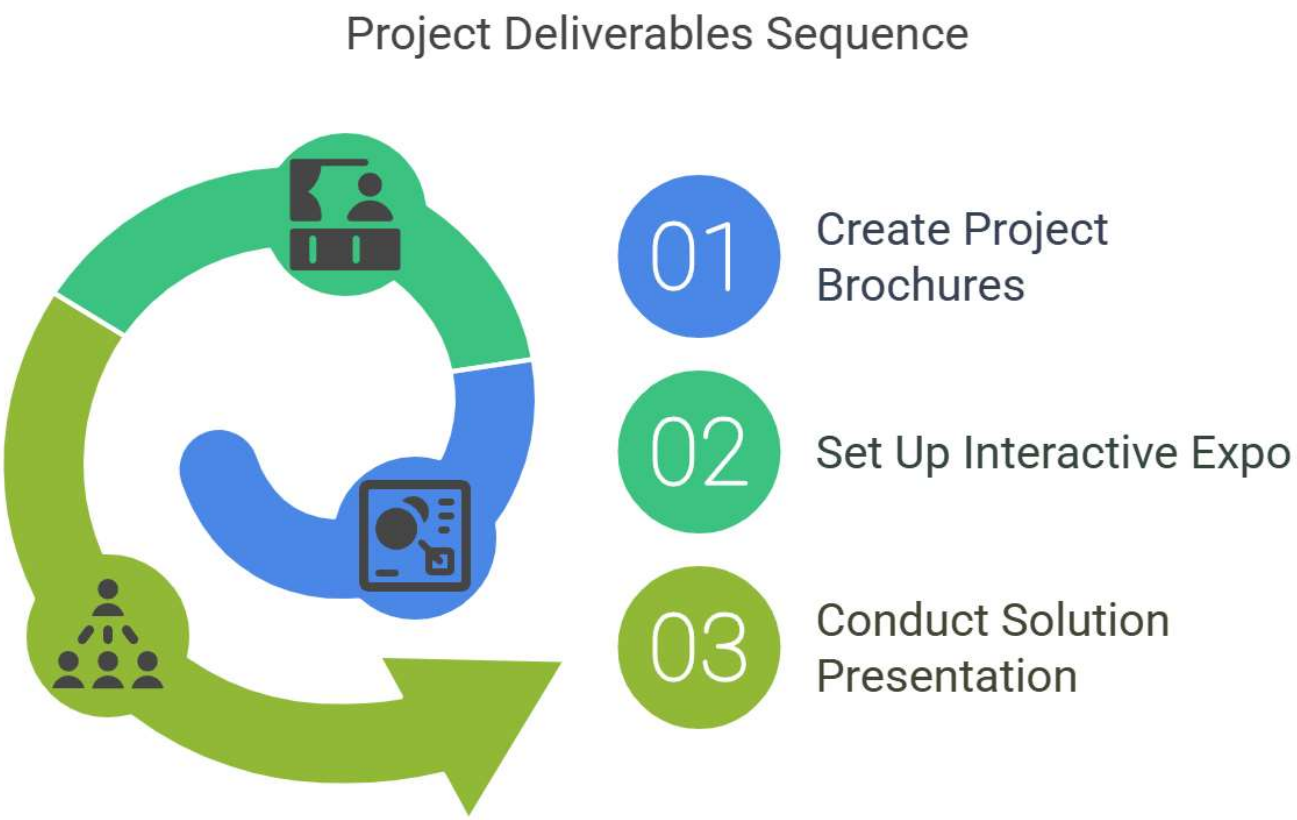
2. Design & Simulation
• Activity:
○ TinkerCAD Prototyping: Design and simulate circuits (light sensors, servo motors, etc).
• Creative Task:
○ Team Logos: Each group creates a logo (a sun + bolt for solar energy).

3. Building & Integration
• Activity:
○ Wooden Hexagon Assembly: Attach Arduino circuits (pre-wired for simplicity) to wooden blocks.
○ Green Hexagons: Add vegetation (fake plants, mini trees) to reserved blocks.
○ Green Monorail: A pre-built elevated rail added to one city as a clean-energy transport example.



4. Communication & Exhibition

- Deliverables:
  - Project Brochures: Summarize solutions with diagrams and QR codes linking to OneNote diaries.
  - Interactive Expo: Visitors activate prototypes (hover hand to turn on solar streetlights).
  - Solution Presentation.



Why Integration Matters

- Integrated Systems Over Individual Projects:
  - Isolated solutions waste resources; connected systems mimic real-world sustainability.
- Scalability: Thematic integration shows how cities can expand efficiently.

Pontos	Nenhum
Enviando	Nada

Entrega	Para	Disponível a partir de	Até
-	Everyone	-	-